

Group Laboratory Report

Lab 4: Experiments on the TCRT5000 Infrared Sensor Module

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Abstract

The TCRT5000 infrared sensor module is widely used in obstacle detection and line-following systems because of its simple structure, low cost, and stable photoelectric performance. In this experiment, the working principle and practical characteristics of the TCRT5000 module were investigated through four experiments, including module function testing, sensitivity adjustment, multi-point distance calibration, and black-line reflection testing. The experiments focused on the relationship between infrared reflection intensity, detection distance, surface color, and sensor output voltage. Experimental results showed that the module could effectively distinguish reflective differences between white and black surfaces, and the detection sensitivity could be adjusted through the onboard potentiometer. Abnormal data behavior at extremely short distances also revealed the nonlinear characteristics and optical limitations of the infrared reflection system. These experiments provided a deeper understanding of the mechanism and practical applications of infrared reflective sensors in embedded electronic systems and intelligent vehicle control systems.

1 Introduction

Infrared reflective sensors are widely applied in modern embedded systems, intelligent vehicles, obstacle avoidance devices, and automatic control systems because they provide non-contact detection with relatively low cost and simple circuitry [1]. Among these sensors, the TCRT5000 infrared sensor module is one of the most commonly used reflective photoelectric sensors. It integrates an infrared emitting diode and a phototransistor into a compact structure, enabling the detection of reflected infrared light from nearby objects [1].

The TCRT5000 module is frequently used in line-following robots and distance detection systems. The sensor determines object characteristics by measuring changes in reflected infrared intensity. White surfaces generally produce stronger infrared reflections, while black surfaces absorb more infrared radiation and therefore produce weaker reflected signals [2]. By combining the infrared sensing structure with an LM393 voltage comparator, the module can provide both analog and digital output signals suitable for microcontroller applications [2].

In practical applications, the performance of the TCRT5000 sensor is influenced by several factors, including detection distance, surface material, ambient light conditions, and sensitivity adjustment. Although the theoretical detection distance is approximately 2–30 mm according to the datasheet [1], the actual detectable range may vary significantly under real experimental conditions. Therefore, it is necessary to experimentally investigate the practical operating characteristics of the sensor module.

In this experiment, four experiments were conducted to study the operational behavior of the TCRT5000 module. The experiments included module functionality verification, sensitivity adjustment through potentiometer rotation, multi-point distance calibration, and black-line reflection testing. Through these experiments, the relationship between reflected infrared intensity and sensor output characteristics was analyzed in detail.

2 Aims of the Experiment

The main aims of this experiment were to understand the internal working principle of the TCRT5000 infrared sensor module and investigate its practical detection characteristics under different experimental conditions. The experiment also aimed to analyze the influence of potentiometer adjustment, object distance, and surface color on infrared reflection intensity and sensor output behavior. In addition, the experiment attempted to evaluate the practical applicability of the module in line-following and obstacle-detection systems.

3 Experimental Observations

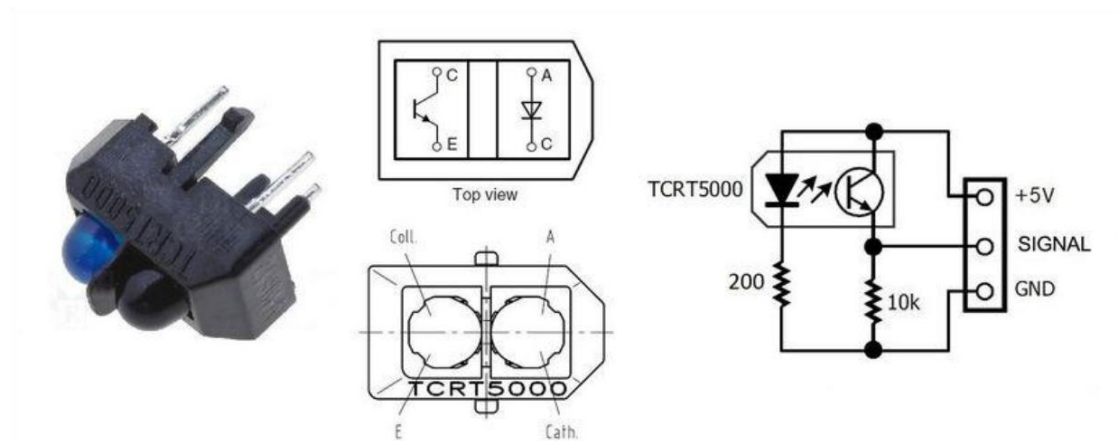
During the experiments, the TCRT5000 module successfully detected reflected infrared light from nearby objects. White paper produced strong reflected infrared signals and caused obvious changes in analog output readings, while black tape produced much weaker reflections. The sensitivity of the module increased when the potentiometer was rotated clockwise, allowing the sensor to detect objects at longer distances. In extremely short-distance conditions, abnormal fluctuations appeared in several

analog output values, indicating that the infrared reflection system may exhibit nonlinear behavior or optical interference effects under certain conditions.

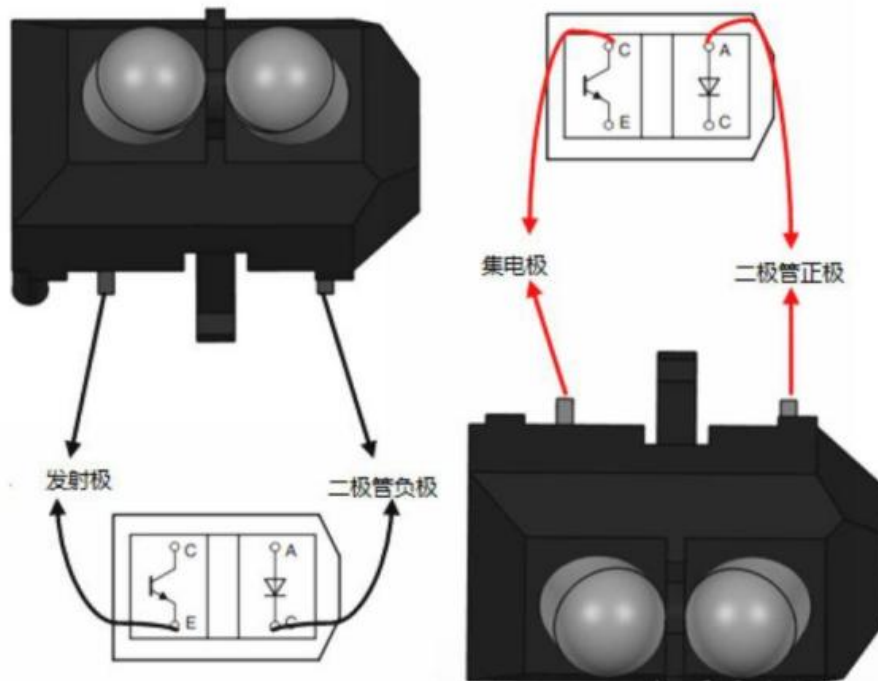
4 Theory and Technical Background

The TCRT5000 infrared sensor module consists of an infrared emitting diode and a phototransistor integrated into a reflective photoelectric sensing structure [1]. The infrared emitting diode continuously emits infrared light, while the phototransistor receives reflected infrared radiation from nearby objects. When sufficient reflected infrared light is detected, the phototransistor conducts and changes the output voltage of the circuit.

The module operates at a voltage range of DC 3V–5V, with 5V recommended for stable operation [1]. In this experiment, the analog output pin A0 was mainly used to observe voltage changes corresponding to infrared reflection intensity.



The module also includes an LM393 voltage comparator for threshold detection and digital signal output [2]. The non-inverting input terminal (+) of the comparator is connected to the sensor signal path, while the inverting terminal (–) is connected to an adjustable potentiometer that controls the reference voltage. When the voltage at the non-inverting terminal is higher than that at the inverting terminal, the comparator output enters a high-impedance state. Conversely, when the non-inverting voltage becomes lower than the inverting voltage, the comparator output switches to a low-level state close to 0V, and the indicator light turns on [2].



The experiments in this study systematically verified the above characteristics of the TCRT5000 module, focusing on its sensitivity, detection range, and response to different surface conditions. Adjusting the potentiometer allowed fine-tuning of the comparator's reference voltage, which directly influenced the sensor's trigger distance and indicator light behavior. Calibration tests across multiple distances confirmed the inverse relationship between detection range and analog output value, as well as the non-linear drop in signal strength beyond a certain threshold. The black tape test further demonstrated the sensor's sensitivity to surface reflectivity, clearly identifying the critical distances at which the reflected signal became too weak to trigger the phototransistor. These results highlight the importance of calibration and environmental factors when using the TCRT5000 module in practical applications. Understanding how distance, color, and sensitivity adjustments affect its performance is essential for optimizing its use in tasks such as line-following robots, obstacle detection, and object counting systems.

5 Experiment 1: Module Function Test

5.1 Experiment Preparation

- (1) **Hardware:** TCRT5000 infrared sensor module, Arduino board, White paper, Ruler.
- (2) **Software:** Arduino IDE.

5.2 Experimental Procedure

Step 1. Connect the TCRT5000 infrared sensor module to the Arduino development board according to the designed circuit. Connect the analog output pin A0 of the sensor module to the analog input port of the Arduino board.

Step 2. Upload the experimental program to the Arduino board and open the serial monitor to observe analog output values in real time.

Step 3. Place a sheet of white paper in front of the sensor and slowly change the distance between the paper and the sensor module.

Step 4. Observe whether the red power indicator and green status indicator light normally during the experiment.

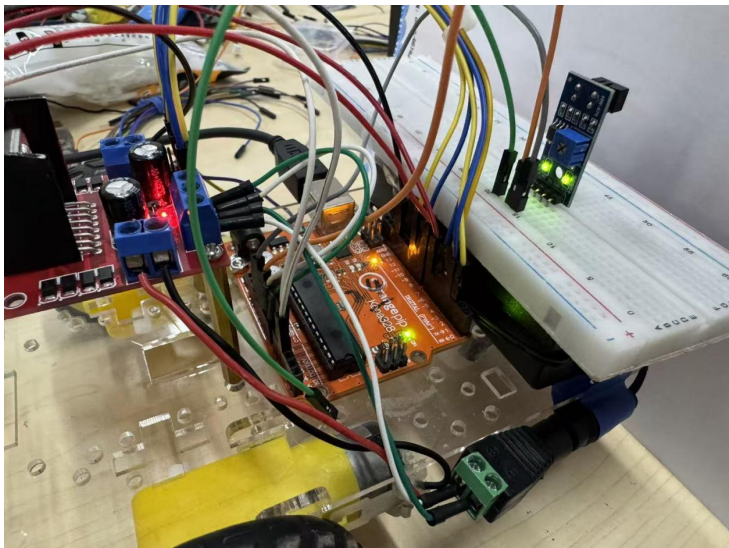
Step 5. Record the analog output values displayed in the serial monitor for each of the three TCRT5000 modules under similar testing conditions.

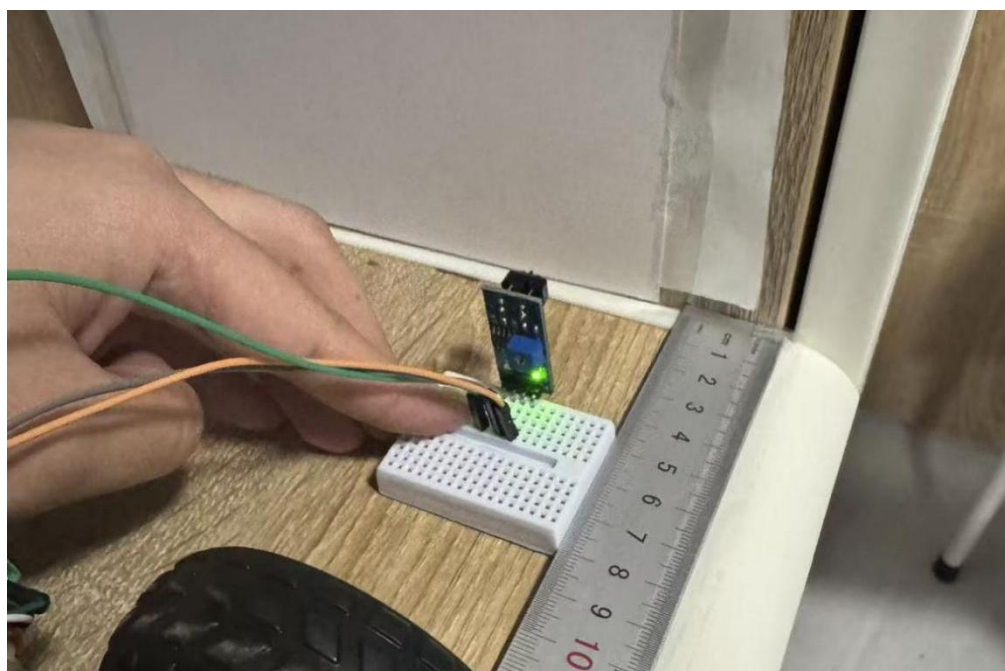
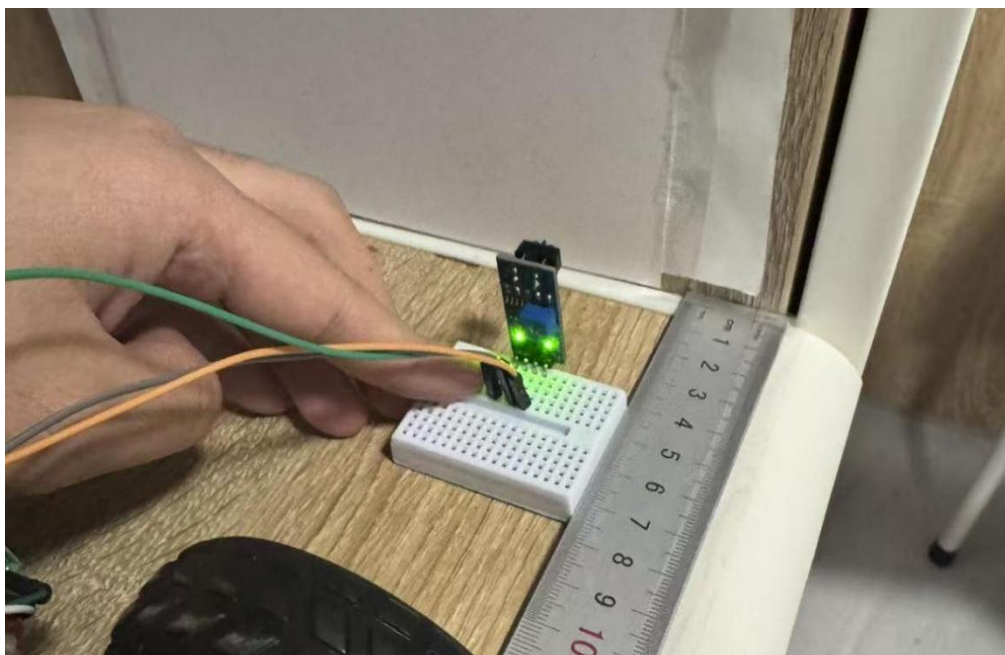
5.3 Relevant Code and Circuit Setup

(1) 5.3.1 Code

```
1  /*Adapted from the example code is in the public domain.
2  http://www.arduino.cc/en/Tutorial/AnalogReadSerial
3  */
4
5  // the setup routine runs once when you press reset:
6  void setup() {
7    // initialize serial communication at 9600 bits per second:
8    Serial.begin(9600);
9  }
10
11 // the loop routine runs over and over again forever:
12 void loop() {
13   // read the input on analog pin 0:
14   int sensorValueLeft = analogRead(A5);
15   delay(2);
16   int sensorValueCentre = analogRead(A4);
17   delay(2);
18   int sensorValueRight = analogRead(A3);
19   delay(2);
20   // print out the value you read:
21   Serial.print("left=");
22   Serial.print(sensorValueLeft);
23   Serial.print("centre=");
24   Serial.print(sensorValueCentre);
25   Serial.print("right=");
26   Serial.println(sensorValueRight);
27   delay(2000); // delay in between reads
28 }
```

(2) 5.3.2 Circuit Setup





5.4 Results

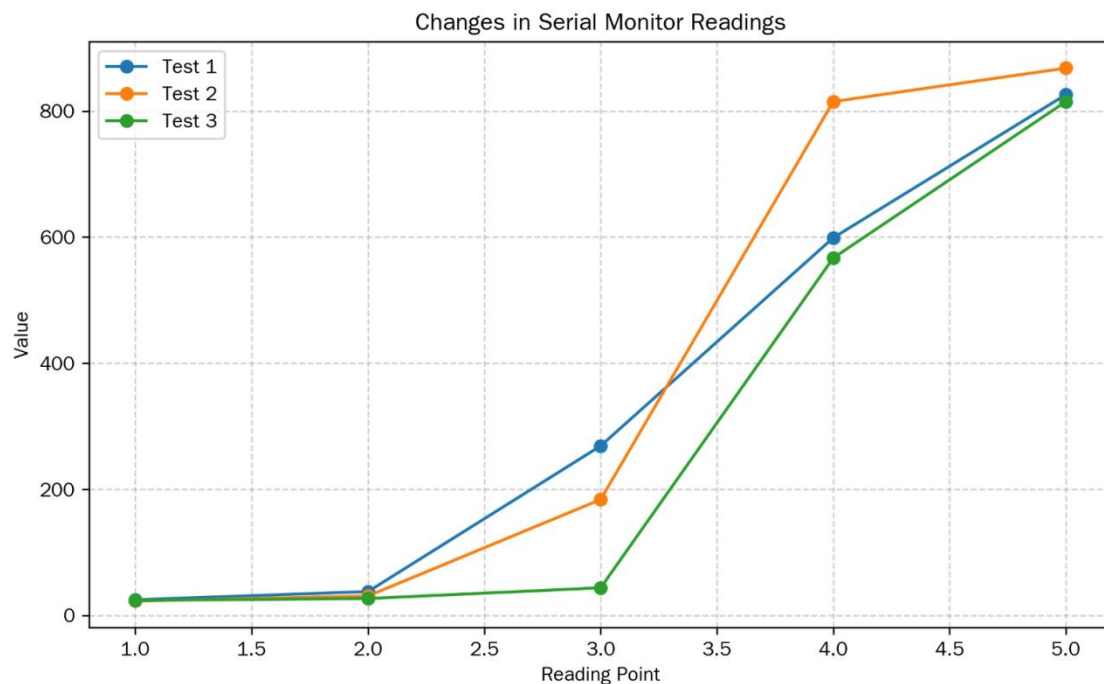
(1) 5.4.1 Experimental results

No.	Changes in serial monitor readings				
1	25	38	269	599	826
2	23	31	184	815	868
3	24	27	44	567	815

(2) 5.4.2 Analysis of Results

In Experiment 1, three TCRT5000 infrared sensor modules were tested using white paper as the reflective target. During the experiment, both the red power indicator and the green status indicator lit normally, indicating that the modules were operating correctly. At the same time, the analog output values from the serial monitor changed significantly when the white paper was moved closer to the sensor.

The recorded data showed that the analog readings increased rapidly after infrared reflection occurred. This demonstrates that white paper has strong infrared reflectivity and can effectively reflect emitted infrared light back to the phototransistor. After receiving sufficient reflected infrared radiation, the phototransistor conducted, resulting in a clear voltage change at the analog output terminal.



Although slight differences existed among the three modules, the overall variation trend remained consistent. These differences were mainly caused by manufacturing tolerances and small variations in infrared emission intensity and phototransistor sensitivity. Overall, the experiment successfully verified that the TCRT5000 module could reliably detect reflected infrared signals and convert them into measurable electrical outputs.

(3) 5.4.3 Error Analysis

Several factors may have affected the experimental results. Ambient light interference could introduce additional infrared radiation into the receiver, causing fluctuations in analog readings. Small differences in the angle and distance between the white paper and the sensor during manual testing may also have influenced the reflection intensity. In addition, slight differences between individual sensor modules may have contributed to the variation observed in the recorded data.

6 Experiment 2: Rotation Direction Impact Test

6.1 Experiment Preparation

- (1) **Hardware:** TCRT5000 infrared sensor module, Arduino board, White paper, Ruler.
- (2) **Software:** Arduino IDE.

6.2 Experimental Procedure

- Step 1.** Use one TCRT5000 infrared sensor module and connect it to the Arduino circuit in the same manner as Experiment 1.
- Step 2.** Upload the program and open the serial monitor for observation.
- Step 3.** Rotate the onboard potentiometer of the sensor module gradually toward the left side and observe the change in detection sensitivity and measurable distance.
- Step 4.** Rotate the potentiometer toward the right side and compare the detection distance and indicator response with the previous condition.
- Step 5.** Record the sensitivity characteristics at the leftmost position, middle position, rightmost position, and final selected test position.
- Step 6.** Avoid rotating the potentiometer completely to the extreme end in order to prevent unstable triggering caused by excessive sensitivity.

6.3 Relevant Code and Circuit Setup

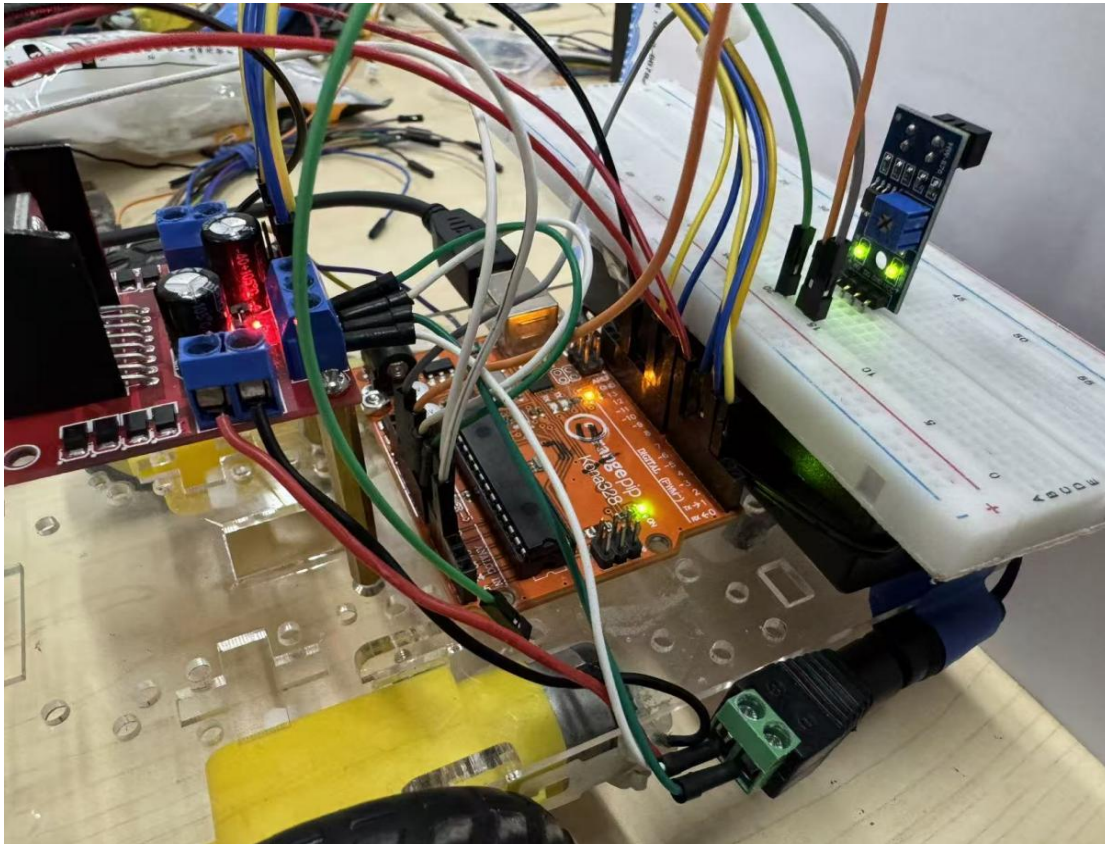
(1) 6.3.1 Code

Same as Experiment 1

```
1  /*Adapted from the example code is in the public domain.
2  http://www.arduino.cc/en/Tutorial/AnalogReadSerial
3  */
4
5  // the setup routine runs once when you press reset:
6  void setup() {
7    // initialize serial communication at 9600 bits per second:
8    Serial.begin(9600);
9  }
10
11 // the loop routine runs over and over again forever:
12 void loop() {
13   // read the input on analog pin 0:
14   int sensorValueLeft = analogRead(A5);
15   delay(2);
16   int sensorValueCentre = analogRead(A4);
17   delay(2);
18   int sensorValueRight = analogRead(A3);
19   delay(2);
20   // print out the value you read:
21   Serial.print("left=");
22   Serial.print(sensorValueLeft);
23   Serial.print("centre=");
24   Serial.print(sensorValueCentre);
25   Serial.print("right=");
26   Serial.println(sensorValueRight);
27   delay(2000);    // delay in between reads
28 }
```

(2) 6.3.2 Circuit Setup

Same as Experiment 1



4.1 6.4 Results

(1) 6.4.1 Experimental results

Position	Leftmost	Middle	Rightmost	Final test position
Sensitivity	Lowest	Medium	Highest	High
Diagram				

(2) 6.4.2 Analysis of Results

In Experiment 2, the onboard potentiometer of the TCRT5000 module was rotated to investigate how

sensitivity adjustment influences the detection range. The results showed that rotating the potentiometer clockwise increased the sensitivity and extended the measurable detection distance. The rightmost rotation direction produced the highest sensitivity among the tested positions.

This behavior can be explained by the operating principle of the LM393 voltage comparator. Rotating the potentiometer changes the reference voltage at the comparator input terminal. When rotated clockwise, the threshold voltage becomes lower, meaning weaker reflected infrared signals can trigger the comparator output. As a result, the sensor becomes capable of detecting objects at greater distances.

However, excessive sensitivity adjustment may also reduce system stability because weak environmental reflections and noise signals can accidentally trigger the sensor. Therefore, the potentiometer was not adjusted completely to the extreme end during the experiment.

(3) 6.4.3 Error Analysis

The potentiometer adjustment was performed manually, making it difficult to ensure identical rotational positions during every test. Environmental infrared interference from surrounding light sources may also have affected the sensitivity measurement. Furthermore, the reflective properties of the target surface and slight changes in sensor alignment could introduce additional experimental deviations.

7 Experiment 3: Multi-Point Distance Calibration

7.1 Experiment Preparation

(1) **Hardware:** TCRT5000 infrared sensor module, Arduino board, White paper, Black tape, Ruler. .

(2) **Software:** Arduino IDE.

7.2 Experimental Procedure

Step 1. Adjust the potentiometer of the TCRT5000 module to the direction corresponding to the larger detection range determined in Experiment 2.

Step 2. Connect the sensor module to the Arduino board and open the serial monitor.

Step 3. Place a white reflective surface in front of the sensor module.

Step 4. Measure the distance between the sensor and the reflective surface using a ruler, beginning from the farthest measurable distance and gradually decreasing the distance.

Step 5. Record the analog output value from the serial monitor at each measurement point.

Step 6. Repeat the measurements for multiple distances and organize the results into a data table for analysis.

7.3 Relevant Code and Circuit Setup

(1) 7.3.1 Code

Same as Experiment 1

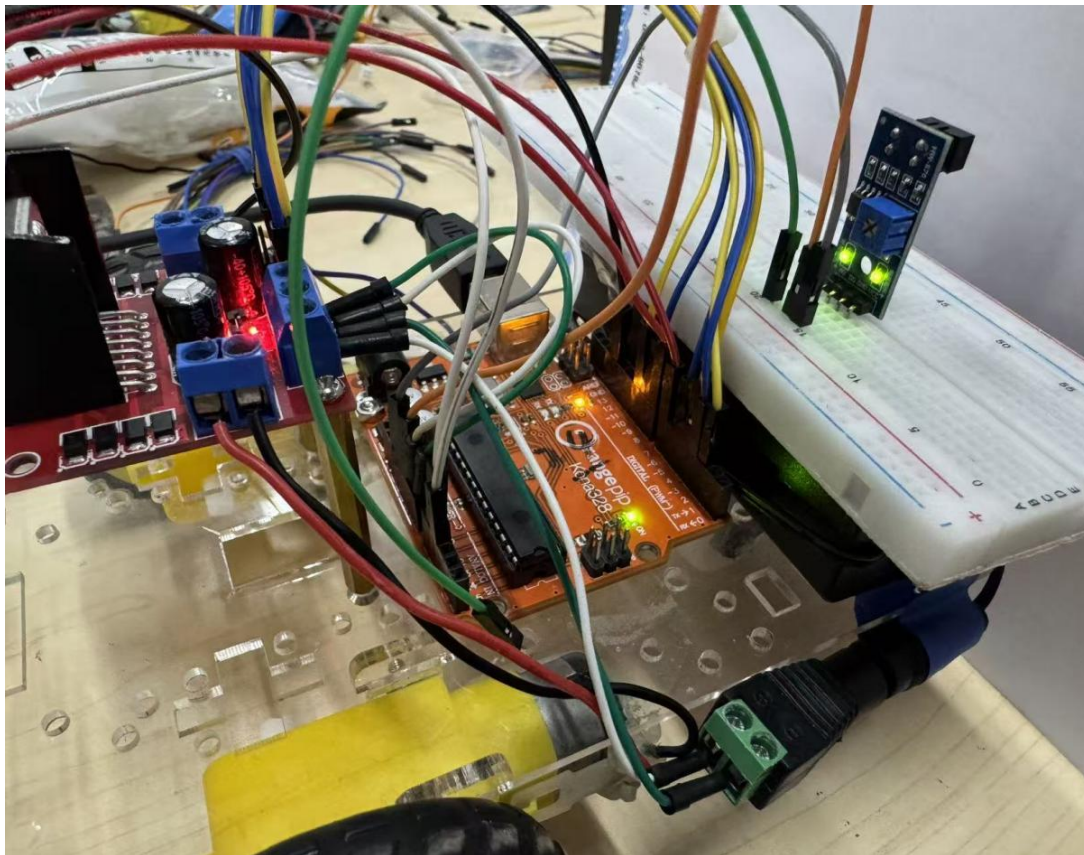

```

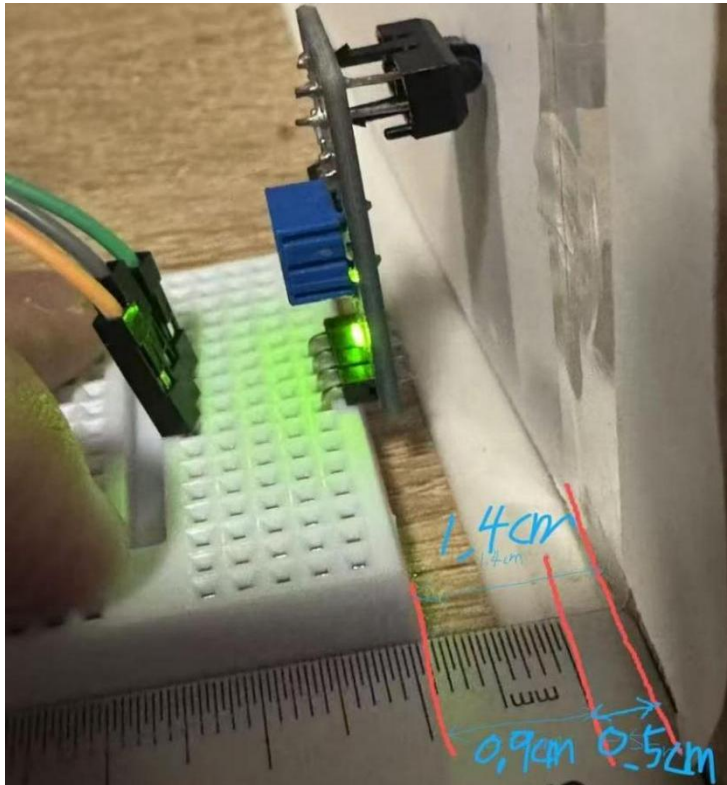
1  /*Adapted from the example code is in the public domain.
2  http://www.arduino.cc/en/Tutorial/AnalogReadSerial
3  */
4
5  // the setup routine runs once when you press reset:
6  void setup() {
7    // initialize serial communication at 9600 bits per second:
8    Serial.begin(9600);
9  }
10
11 // the loop routine runs over and over again forever:
12 void loop() {
13   // read the input on analog pin 0:
14   int sensorValueLeft = analogRead(A5);
15   delay(2);
16   int sensorValueCentre = analogRead(A4);
17   delay(2);
18   int sensorValueRight = analogRead(A3);
19   delay(2);
20   // print out the value you read:
21   Serial.print("left=");
22   Serial.print(sensorValueLeft);
23   Serial.print("centre=");
24   Serial.print(sensorValueCentre);
25   Serial.print("right=");
26   Serial.println(sensorValueRight);
27   delay(2000);    // delay in between reads
28 }

```

(2) 7.3.2 Circuit Setup

Same as Experiment 1





7.4 Results

(1) 7.4.1 Experimental results

d : Actual distance from the infrared emitter to the vertical paper plane (unit: cm)

R : Breadboard right-edge reading (unit: cm)

L_s : Scale margin length, fixed at 0.5 cm

L_m : Module width, fixed at 1.4 cm

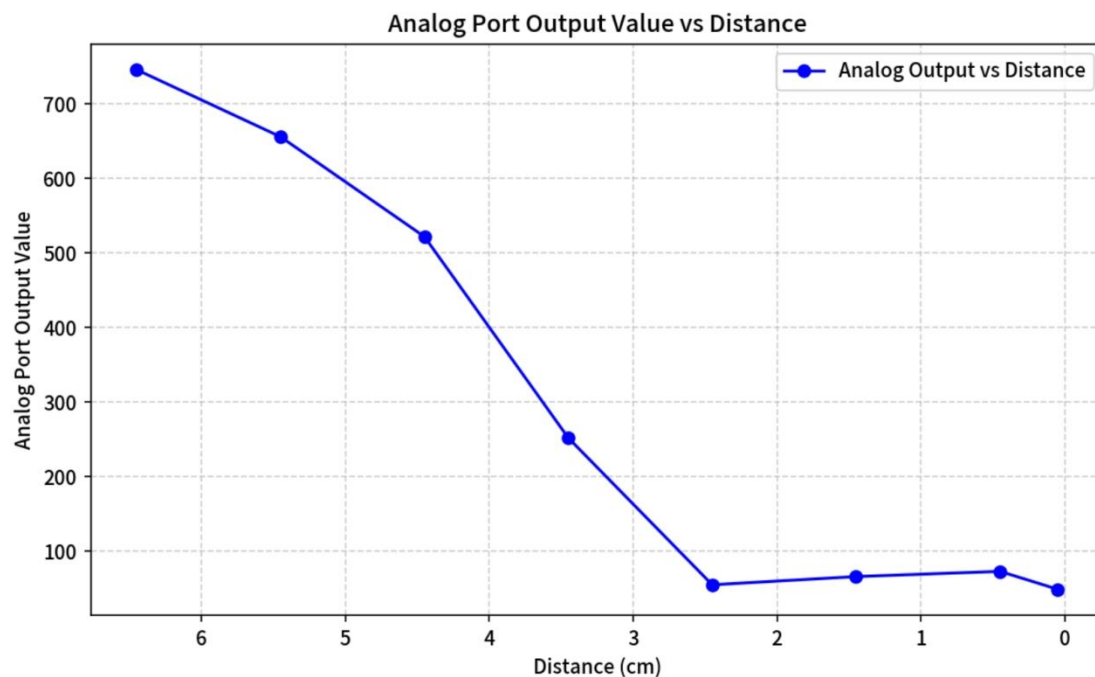
No.	Distance (cm)	Analog port output value
1	6.45	746
2	5.45	656
3	4.45	522
4	3.45	252
5	2.45	55
6	1.45	66
7	0.45	73
8	0.05	49

(2) 7.4.2 Analysis of Results

In Experiment 3, the potentiometer was adjusted to the direction corresponding to the larger detection range, and the analog output values were measured at multiple distances. The overall experimental trend showed that the analog output value generally decreased as the distance between the sensor and the reflective object decreased.

This phenomenon is related to the conduction characteristics of the phototransistor and the voltage distribution of the sensing circuit. As reflected infrared intensity increases, the phototransistor conducts more strongly, causing the output voltage at the analog terminal to decrease.

However, abnormal behavior appeared in the seventh and eighth groups of data. According to the expected trend, the analog output value should continue decreasing as the sensor approaches the reflective surface. Instead, the measured values increased unexpectedly at extremely short distances.



This abnormal phenomenon may be explained by several factors. First, when the reflective surface becomes extremely close to the sensor, the infrared emitter and receiver may no longer satisfy the optimal reflection geometry. Because the TCRT5000 emitter and receiver are physically separated, reflected infrared light may partially bypass the receiving path at extremely short distances. Second, excessive reflected infrared intensity may cause partial saturation or nonlinear behavior of the phototransistor, resulting in unstable voltage output. Third, strong reflections at very short distances may introduce optical scattering and irregular reflection angles, further affecting the received signal intensity.

Therefore, although the general relationship between distance and analog output remained valid, the experiment demonstrated that the TCRT5000 infrared sensing system does not behave linearly under extremely short-distance conditions.

(3) 7.4.3 Error Analysis

The distance intervals in this experiment were very small, so tiny measurement errors could

significantly affect the final readings. Hand movement during measurement may also have slightly changed the reflection angle between the sensor and the target surface. In addition, analog signal noise and environmental infrared interference may have contributed to fluctuations in the recorded values, especially at extremely short distances where the sensor output becomes highly sensitive.

8 Experiment 4: Black Tape Reflection Threshold Test

8.1 Experiment Preparation

(1) **Hardware:** TCRT5000 infrared sensor module, Arduino board, White paper, Black tape, Ruler. .

(2) **Software:** Arduino IDE.

8.2 Experimental Procedure

Step 1. Attach a strip of black tape with a width of approximately 2 cm onto a sheet of white paper to simulate a line-following environment.

Step 2. Connect the TCRT5000 sensor module to the Arduino circuit and upload the experimental program.

Step 3. Place the sensor above the black tape and gradually adjust the distance between the sensor and the paper surface.

Step 4. Observe the analog output values and the status of the indicator light during the distance adjustment process.

Step 5. Record the distance at which reflected infrared light first becomes detectable and the distance at which the reflected signal disappears.

8.3 Relevant Code and Circuit Setup

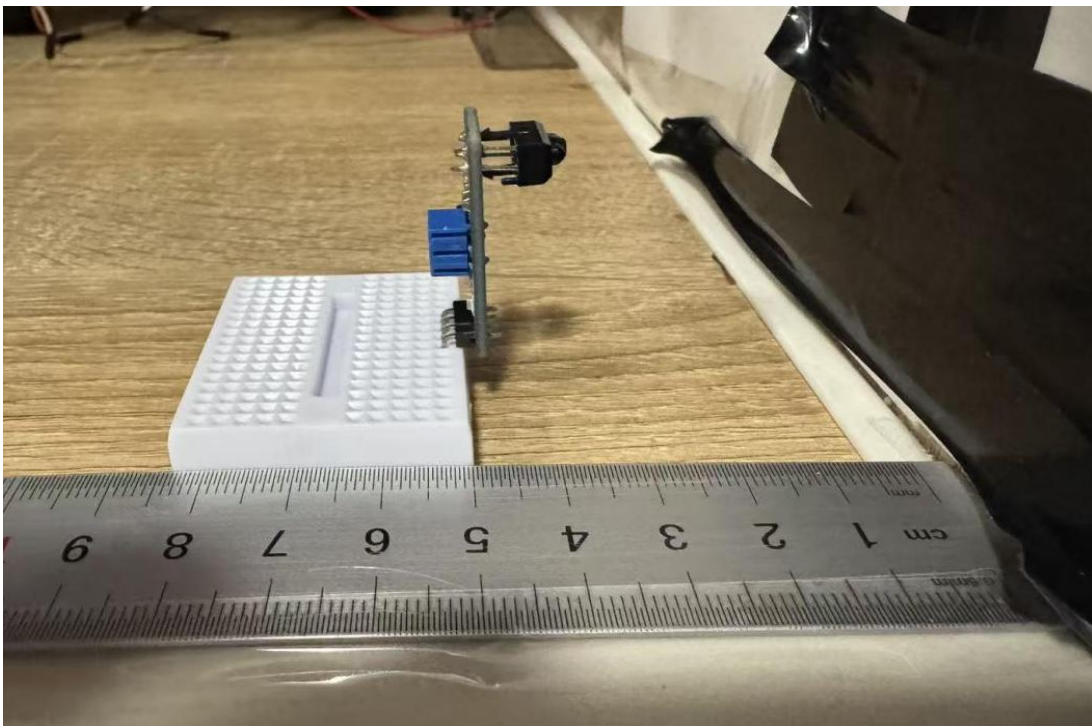
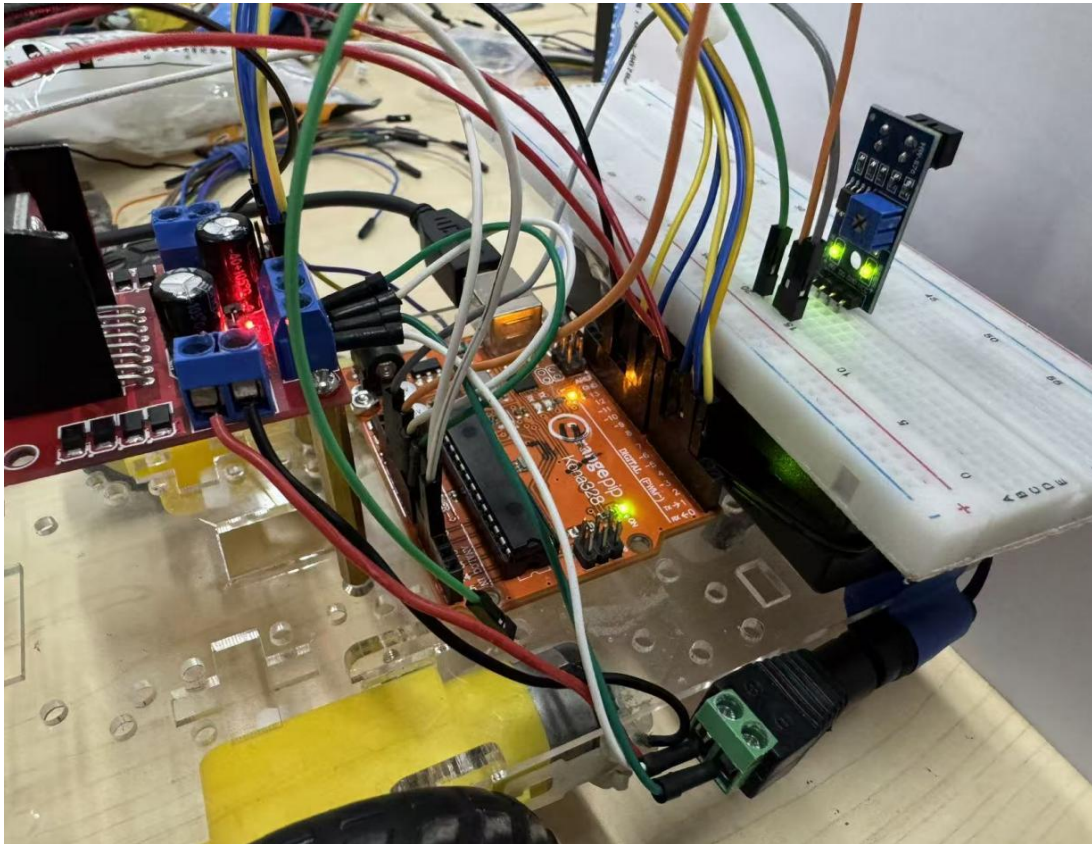
(1) 8.3.1 Code

Same as Experiment 1

```
1  /*Adapted from the example code is in the public domain.
2  http://www.arduino.cc/en/Tutorial/AnalogReadSerial
3  */
4
5  // the setup routine runs once when you press reset:
6  void setup() {
7    // initialize serial communication at 9600 bits per second:
8    Serial.begin(9600);
9  }
10
11 // the loop routine runs over and over again forever:
12 void loop() {
13   // read the input on analog pin 0:
14   int sensorValueLeft = analogRead(A5);
15   delay(2);
16   int sensorValueCentre = analogRead(A4);
17   delay(2);
18   int sensorValueRight = analogRead(A3);
19   delay(2);
20   // print out the value you read:
21   Serial.print("left=");
22   Serial.print(sensorValueLeft);
23   Serial.print("centre=");
24   Serial.print(sensorValueCentre);
25   Serial.print("right=");
26   Serial.println(sensorValueRight);
27   delay(2000); // delay in between reads
28 }
```


(2) 8.3.2 Circuit Setup

Same as Experiment 1



8.4 Results

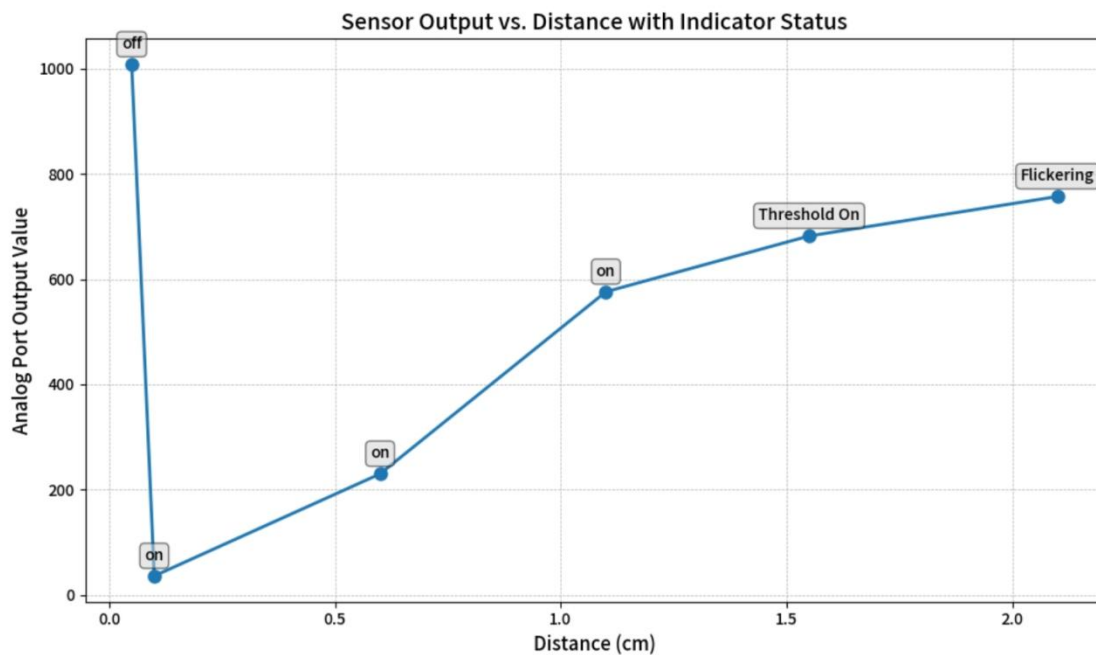
(1) 8.4.1 Experimental results

No.	Distance (cm)	Analog port output value	Indicator light status
1	0.05	1008	off
2	0.1	36	on
3	0.6	230	on
4	1.1	576	on
5	1.55	682	Threshold On
6	2.1	757	Flickering

(2) 8.4.2 Analysis of Results

In Experiment 4, black tape with a width of approximately 2 cm was attached to white paper in order to simulate a line-following environment. The experiment investigated the critical distance at which reflected infrared light from the black surface could still be detected by the sensor.

The results showed that at extremely short distances, the indicator light remained off because the black tape absorbed most of the emitted infrared radiation and produced very weak reflected signals. As the distance increased slightly, the indicator light turned on and the analog output value changed significantly, indicating that sufficient reflected infrared light was reaching the phototransistor.



When the sensor distance increased further, the reflected infrared intensity gradually weakened. Near the threshold distance, the indicator light began flickering, indicating that the comparator output was repeatedly switching between high and low states due to unstable reflection intensity. This

phenomenon demonstrates that the TCRT5000 module is highly sensitive to changes in surface reflectivity and distance.

The experiment also verified the basic operating principle of line-following robots. White surfaces reflect stronger infrared signals, while black surfaces absorb infrared radiation more effectively. The sensor can therefore distinguish black lines from white backgrounds based on differences in reflected infrared intensity.

(3) 8.4.3 Error Analysis

The reflection characteristics of the black tape may have been influenced by surface roughness and material properties. Slight changes in sensor height or tilt angle could significantly affect the received infrared intensity near the threshold region. Environmental lighting conditions and electrical noise may also have contributed to fluctuations in the analog readings and caused indicator light flickering.

9 Conclusion

In this experiment, the working principles and practical operating characteristics of the TCRT5000 infrared sensor module were systematically investigated through four independent experiments. The experiments verified that the module can effectively detect reflected infrared light and convert optical signals into measurable electrical outputs.

The experimental results demonstrated that the output characteristics of the sensor are closely related to object distance, surface color, reflection intensity, and sensitivity adjustment. White surfaces produced strong reflected infrared signals, while black surfaces absorbed infrared radiation and generated much weaker reflections. By rotating the onboard potentiometer clockwise, the sensitivity and measurable detection range of the module increased significantly.

The experiments also revealed that the infrared reflection system may exhibit nonlinear behavior under extremely short-distance conditions. Abnormal output changes observed in several data groups were likely caused by optical geometry limitations, phototransistor saturation, and irregular reflection paths. In addition, threshold flickering observed during black-line testing demonstrated the sensitivity of the LM393 comparator to weak infrared signals.

Overall, the experiments provided a deeper understanding of the TCRT5000 infrared sensor module and its applications in obstacle detection, line-following robots, and embedded electronic systems.

10 References

- [1] Vishay Semiconductors, “TCRT5000 Reflective Optical Sensor Datasheet,” Vishay Intertechnology, 2024.
- [2] Texas Instruments, “LM393 Low Power Low Offset Voltage Dual Comparators Datasheet,” Texas Instruments Incorporated, 2023.
- [3] Arduino, “Analog Read Serial,” Arduino Documentation.
- [4] A. Malvino and D. Bates, Electronic Principles, 8th ed. New York, NY, USA: McGraw-Hill, 2015.
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